25 Here are the first six terms of an arithmetic sequence.

3 8 13 18 23 28

(a) Find an expression, in terms of n, for the nth term of this sequence.

The common difference is +5, so start with 5n.

5n gives 5, 10, 15, 20, ...

Subtract 2 each time: nth term = 5n – 2.

2 marks total
Correct nth term

5n – 2

(2)

The nth term of a different sequence is $3n^2$
Nathan says that the 4th term of this sequence is 144

(b) Is Nathan right?
Show how you get your answer.

For the 4th term, substitute n = 4.

$3 \times 4^2 = 3 \times 16 = 48$

No. The 4th term is 48, not 144.

(1)

1 mark
No, supported by correct substitution

(Total for Question 25 is 3 marks)

TOTAL FOR PAPER IS 80 MARKS

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